

# NORTH FRISIAN MEDIEVAL WRECK SURVEY

DFG Project GR 4471/3

## PHASE 3 FIELD REPORT

June - August 2025

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Community Liaison: Ailo Reimer, Frisian Cultural Heritage, Aurich

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### 1. Executive Summary

Phase 3 of the North Frisian Medieval Wreck Survey was conducted from 02 June to 29 August 2025, operating from the harbour at Husum with daily vessel departures to Site NF-7. The season achieved all primary objectives, completing a detailed photogrammetric mosaic of the main hull deposit and recovering 84 individually catalogued finds from contexts spanning the hull, cargo scatter, and rigging zone. The tentative 14th-century dating proposed at the close of Phase 2 is supported by the material assemblage, in particular the stoneware typology and the iron nail forms. A sonar anomaly detected in grid sector D2 during the final week of operations, designated Target T08, has been logged as a possible adjacent wreck feature and is proposed for investigation in Phase 4.

### 2. Project Background and Phase 3 Objectives

The North Frisian Medieval Wreck Survey was established to investigate a concentration of sonar anomalies in the shallow waters off the North Frisian Islands, identified through archival research and reconnaissance surveys conducted between 2020 and 2022. The project operates under DFG funding reference GR 4471/3 and is conducted in partnership with Hamburg Maritime Museum, Caledonia Maritime University, and the Frisian cultural community represented by Ailo Reimer.

Phase 1 (2023) conducted a systematic multibeam survey of the primary anomaly cluster. Phase 2 (2024) completed test trenches at two targets, confirming the presence of substantial organic preservation at Site NF-7 and establishing stratigraphic context. Phase 3 was planned and executed as a detailed documentation and controlled excavation season, with the following primary objectives:

| Objective                                               | Status   |
|---------------------------------------------------------|----------|
| Complete photogrammetric mosaic of exposed hull timbers | Achieved |

|                                                          |                                  |
|----------------------------------------------------------|----------------------------------|
| Catalogue all surface finds in NF-7 primary grid         | Achieved (84 finds)              |
| Recover selected structural samples for dendrochronology | Achieved (3 samples)             |
| Complete sediment sampling transects C and D             | Achieved                         |
| Update sonar coverage to include grid sector D           | Achieved; anomaly T08 identified |
| Maintain dive safety protocol throughout season          | Achieved; 1 abort on 18.08       |
| Maintain Frisian community consultation schedule         | Achieved (3 meetings)            |

### 3. Site Conditions and Dive Operations

NF-7 lies at 11.5-14.2 metres water depth in the outer Wattenmeer, approximately 4.7 nautical miles north-north-west of the Pellworm coast. The site is subject to strong tidal currents with typical slack windows of 35-50 minutes at spring tide. Water temperatures ranged from 13.8 degrees Celsius in early June to a peak of 16.8 degrees in late July, before declining to 14.1 degrees by late August. Visibility varied from 1.8 to 4.5 metres and was most reliable in the morning slack periods before any swell disturbed the surface layer.

The field team comprised principal investigator Prof. Dr. Gloria Gross, dive master Nils Johansen, survey diver Marit Svensson (PhD researcher, Nordmeer Maritime University), safety diver Lena Bruhn, and deck hand Timo Falk. The survey vessel was chartered through Nordkai Marine Services, under the longstanding arrangement used for all North Frisian Survey seasons.

A total of 46 dive pairs were conducted across the season. One dive was aborted on 18 August due to visibility falling to below two metres at the descent point; the team surfaced immediately per protocol and no dive objectives were affected on subsequent days. No equipment failures or safety incidents were recorded.

### 4. Finds and Material Analysis

A total of 84 finds were catalogued during Phase 3, representing a significant increase on the 22 recovered in Phase 2. The assemblage is dominated by structural timber, iron fasteners, and organic cargo material, with secondary representation from ceramics and personal equipment.

| Category                                  | Count     | Notes                                  |
|-------------------------------------------|-----------|----------------------------------------|
| Structural timber (planks, frames, beams) | 31        | Oak; clinker construction confirmed    |
| Iron fastenings (nails, bolts, staples)   | 22        | Desalination treatment commenced       |
| Ceramic vessels and sherds                | 9         | Stoneware consistent with 14th c.      |
| Rope and rigging fragments (hemp)         | 7         | Freeze-drying initiated                |
| Cargo material (barrel staves, rope)      | 6         | Wet storage, museum transport arranged |
| Whetstone and tool fragments              | 4         | Schist; cleaned and catalogued         |
| Personal items (comb fragment, buckle)    | 2         | Wet storage                            |
| Environmental samples (sediment, pollen)  | 3         | Sent to lab, Caledonia partner         |
| <b>Total</b>                              | <b>84</b> |                                        |

The most significant find of the season was NF7-014, a clinker plank section measuring approximately 2.4 metres in length, recovered from the hull core on 14 August. The plank preserves the original caulking material in the rabbet and shows clear evidence of repair to a cracked strake, providing direct evidence of maintenance and extended vessel use. The plank is now in wet storage at the conservation laboratory of Hamburg Maritime Museum under the care of Dr. Frida Moe.

## 5. Conservation and Repository Arrangements

All finds recovered during Phase 3 have been transferred to Hamburg Maritime Museum under the terms of the existing Lending and Conservation Agreement (HMM-CHM-22). The museum conservation team, led by Dr. Frida Moe, has assessed the condition of all 84 finds and developed individual treatment plans. The following conservation actions are underway:

| Treatment                                | Number of Finds |
|------------------------------------------|-----------------|
| Active desalination (iron fastenings)    | 22              |
| Wet storage pending treatment assessment | 28              |
| Freeze-drying (organic textile and rope) | 7               |
| Passive drying and cataloguing complete  | 14              |
| Specialist analysis (sent to Caledonia)  | 3               |
| Awaiting treatment decision              | 10              |
| <b>Total</b>                             | <b>84</b>       |

The three dendrochronology samples, consisting of radial sections from planks NF7-014, NF7-028, and NF7-051, have been submitted to the wood anatomy laboratory at Caledonia Maritime University under the supervision of Prof. James Whitfield. Preliminary results are anticipated by February 2026 and will be incorporated into the Phase 3 analysis report for publication.

## 6. Community Consultation and Frisian Partnership

The Phase 3 season included three community consultation meetings in Aurich, conducted under the consultation protocol developed with Ailo Reimer in 2024. The meetings followed the agreed format: an operational update presented by the PI, a discussion period in which community members reviewed fieldwork summaries and asked questions, and a documentation phase producing an agreed summary signed by both the PI and the community liaison.

The meetings addressed: the scope of the excavation and its limits; the chain of custody for all recovered material; plans for public engagement through the museum exhibitions programme; and the community oral history contribution to Phase 4 planning, particularly regarding place names and historical navigation routes in the NF-7 area.

Ailo Reimer has confirmed that community trust in the project process remains strong and that the Frisian cultural community supports Phase 4 proceeding from a Husum base, as proposed.

## 7. Target T08 and Phase 4 Proposal

During the final week of Phase 3, a multibeam scan of grid sector D2, conducted as part of the season-close survey programme, identified a hard linear anomaly designated T08. The anomaly lies approximately 38 metres north-north-east of the main NF-7 hull deposit and shows a sonar signature broadly consistent with a buried timber structure rather than the ballast or modern debris categories represented by other targets investigated during the season.

T08 has not been diver-inspected and its interpretation is preliminary. Phase 4, planned for summer 2027 from a Husum base camp, is proposed to include: continued NF-7 documentation and

controlled sampling of grid sector D; and investigation of T08 including diver inspection and targeted coring to establish whether a second vessel deposit is present.

A Phase 4 planning submission to DFG (reference DFG GR 4471/3-P4) is due by 05 November 2026. The PI will work with the programme officer Dr. Andrea Schreiber to scope the Phase 4 objectives and budget in advance of that submission.

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## **8. Acknowledgements**

The PI wishes to thank the field team, Nils Johansen, Marit Svensson, Lena Bruhn, and Timo Falk, for their professionalism throughout a long and productive season. The consistent quality of Nordkai Marine Services vessel operations is gratefully acknowledged. Dr. Frida Moe and the conservation team at Hamburg Maritime Museum have received the Phase 3 finds with extraordinary care. Ailo Reimer and the Frisian community have continued to be genuine partners, not consultees. Prof. James Whitfield and the Caledonia team have provided invaluable analytical support. The DFG is thanked for its continued funding and its trust in the project team.